

CG Infinity - Git & GitHub Hands-on Assignment

Project: Employee Management System

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Task 1

Objective: Create a new project folder named EmployeeManagementSystem and initialize it as a Git repository.

```
C:\Users\ARTH SHARMA>cd EmployeeManagementSystem

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git init
Reinitialized existing Git repository in C:/Users/ARTH SHARMA/EmployeeManagementSystem/.git/

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

Task 2

Objective: Create README.md, index.html and style.css. Add project title and description in README.md.

```
C:\Users\ARTH SHARMA\EmployeeManagementSystem>type nul >README.md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>type nul > index.html

C:\Users\ARTH SHARMA\EmployeeManagementSystem>type nul >style.css

C:\Users\ARTH SHARMA\EmployeeManagementSystem>dir
Volume in drive C is Windows
Volume Serial Number is DC68-721D

Directory of C:\Users\ARTH SHARMA\EmployeeManagementSystem

22-06-2026  00:23    <DIR>          .
22-06-2026  00:04    <DIR>          ..
22-06-2026  00:23                0 index.html
22-06-2026  00:23                0 Readme.md
22-06-2026  00:23                0 style.css
                3 File(s)                0 bytes
                2 Dir(s)  33,242,787,840 bytes free

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git status
On branch master
```

Task 3

Objective: Verify repository status and identify untracked files.

```
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md
        index.html
        style.css

nothing added to commit but untracked files present (use "git add" to track)
```

Task 4

Objective: Add files to staging area and create the first commit.

```
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add .

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   Readme.md
        new file:   index.html
        new file:   style.css

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Initial project setup"
[master (root-commit) ca17ed3] Initial project setup
 3 files changed, 6 insertions(+)
 create mode 100644 Readme.md
 create mode 100644 index.html
 create mode 100644 style.css
```

Task 5

Objective: Review repository history using git log.

```
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git log
commit ca17ed36077ec4145f42b9c26821008d5aa6de21 (HEAD -> master)
Author: ARTHSHARMA58 <arth89408@gmail.com>
Date:   Mon Jun 22 00:28:59 2026 +0530

    Initial project setup

C:\Users\ARTH SHARMA\EmployeeManagementSystem>
```

Task 6

Objective: Create feature/employee-registration branch and switch to it.

```

Initial project setup

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git branch featue/employee-registration

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch featue/employee_registration
fatal: invalid reference: featue/employee_registration

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch featue/employee-registration
Switched to branch 'featue/employee-registration'

C:\Users\ARTH SHARMA\EmployeeManagementSystem>

```

Task 7

Objective: Create employee-registration.html and commit changes.

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>notepad employee-registration.html

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git status
On branch featue/employee-registration
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    employee-registration.html

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Added employee registration page"
On branch featue/employee-registration
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    employee-registration.html

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add employee-registration.html

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Added employee registration page"
[featue/employee-registration 2881973] Added employee registration page
1 file changed, 9 insertions(+)
create mode 100644 employee-registration.html

```

Task 8

Objective: Compare working directory changes with last committed version using git diff.

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>notepad employee-registration.html

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git diff
diff --git a/employee-registration.html b/employee-registration.html
index ffd5033..c02345c 100644
--- a/employee-registration.html
+++ b/employee-registration.html
@@ -5,5 +5,6 @@
 </head>
 <body>
   <h1>Employee Registration Form</h1>
+  <h2>Register New Employee</h2>
 </body>
</html>
\ No newline at end of file

```

Task 9

Objective: Merge feature/employee-registration branch into main/master.

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch master
Switched to branch 'master'

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git merge featue?employee-registration
merge: featue?employee-registration - not something we can merge

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git merge featue/employee-registration
Updating ca17ed3..59a0388
Fast-forward
 employee-registration.html | 10 ++++++++
 1 file changed, 10 insertions(+)
 create mode 100644 employee-registration.html

```

Task 10

Objective: Create GitHub repository and connect local repository.

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git remote add origin https://github.com/ARTHSHARMA58/EmployeeManagementSystem

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git remote -v
origin  https://github.com/ARTHSHARMA58/EmployeeManagementSystem (fetch)
origin  https://github.com/ARTHSHARMA58/EmployeeManagementSystem (push)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>

```

Task 11

Objective: Publish project to GitHub.

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git push -u origin master
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (10/10), 973 bytes | 324.00 KiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/ARTHSHARMA58/EmployeeManagementSystem
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

```

Task 12

Objective: Create feature/employee-dashboard branch and commit dashboard page.

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git branch feature/employee-dashboard

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch feature/employee-dashboard
Switched to branch 'feature/employee-dashboard'

C:\Users\ARTH SHARMA\EmployeeManagementSystem>type nul > employee-dashboard.html

C:\Users\ARTH SHARMA\EmployeeManagementSystem>notepad employee-dashboard.html

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add .

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m"Added employee dashboard page"
[feature/employee-dashboard bffb2a3] Added employee dashboard page
 1 file changed, 9 insertions(+)
 create mode 100644 employee-dashboard.html

```

Task 13

Objective: Synchronize local repository with latest GitHub changes.

```
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git pull origin master
From https://github.com/ARTHSHARMA58/EmployeeManagementSystem
* branch      master      -> FETCH_HEAD
Already up to date.
```

Task 14

Objective: Merge dashboard feature into main/master branch.

```
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git merge feature/employee-dashboard
Updating 59a0388..bffb2a3
Fast-forward
 employee-dashboard.html | 9 ++++++++
 1 file changed, 9 insertions(+)
 create mode 100644 employee-dashboard.html

C:\Users\ARTH SHARMA\EmployeeManagementSystem>dir
Volume in drive C is Windows
Volume Serial Number is DC68-721D

Directory of C:\Users\ARTH SHARMA\EmployeeManagementSystem

22-06-2026  00:52    <DIR>          .
22-06-2026  00:51    <DIR>          ..
22-06-2026  00:52             138 employee-dashboard.html
22-06-2026  00:44             181 employee-registration.html
22-06-2026  00:23              0 index.html
22-06-2026  00:26             117 Readme.md
22-06-2026  00:23              0 style.css
                5 File(s)              436 bytes
                2 Dir(s) 33,222,971,392 bytes free
```

Task 15

Objective: Display all branches and identify active branch.

```
2 Dir(s) 33,222,971,392 bytes free

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git branch
  featue/employee-registration
  feature/employee-dashboard
* master
```

Task 16

Objective: Create and resolve merge conflict between login and dashboard branches.

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch feature/login
M      Readme.md
Switched to branch 'feature/login'

C:\Users\ARTH SHARMA\EmployeeManagementSystem>notepad README.md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add README.md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Updated README for login feature"
On branch feature/login
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readme.md

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch master
M      Readme.md
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch feature employee-dashboard
fatal: only one reference expected

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch feature/employee-dashboard
M      Readme.md
Switched to branch 'feature/employee-dashboard'

C:\Users\ARTH SHARMA\EmployeeManagementSystem>notepad README .md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>notepad README.md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add README.md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Updated README for dashboard feature"
On branch feature/employee-dashboard
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)

```

```

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add .

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Login feature update"
[login 962288b] Login feature update
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git branch dashboard

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch dashboard
Switched to branch 'dashboard'

C:\Users\ARTH SHARMA\EmployeeManagementSystem>notepad README.md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add .

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Dashboard feature update"
[dashboard 918ef1a] Dashboard feature update
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git switch master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git merge login
Updating bffb2a3..962288b
Fast-forward
  Readme.md | 2 +--
  1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git merge dashboard
Auto-merging Readme.md
CONFLICT (content): Merge conflict in Readme.md
Automatic merge failed; fix conflicts and then commit the result.

C:\Users\ARTH SHARMA\EmployeeManagementSystem>

```

```
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SHARMA
# Employee Management System - Login & Dashboard Features

SHARMA

SHARMA
A simple web-based Employee Management System for managing employee records.

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```

```
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add README.md
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git status
On branch master
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git add Readme.md
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git status
On branch master
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)

All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   Readme.md

C:\Users\ARTH SHARMA\EmployeeManagementSystem>
C:\Users\ARTH SHARMA\EmployeeManagementSystem>git commit -m "Resolved merge conflict"
[master 78ad4d6] Resolved merge conflict

C:\Users\ARTH SHARMA\EmployeeManagementSystem>git diff
C:\Users\ARTH SHARMA\EmployeeManagementSystem>
```